THE SETS OF POINTS OF ALGEBRAIC STACKS CARRY A UNIQUE TOPOLOGY CONTINUOUS FOR EVERY MORPHISM AND OPEN FOR FLAT LFP MAPS FROM SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1154-1165 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 1167-1252, the complete author proof (84 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 0 of queue_stacks-properties.json (pb=3301, stmt 1154-1165, proof 1167-1252).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-topology-points' also exists in stacks-spaces-properties — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04T1: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1252–1252. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-points-cartesian: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1019–1036. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-property-spaces-too: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 650–658. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03ME: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1252–1252. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 042S: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1252–1252. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04TA: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1252–1252. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04AJ: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1252–1252. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *There exists a unique topology on the sets of points of algebraic stacks with the following properties:*

1. *for every morphism of algebraic stacks $\mathcal{X} \to \mathcal{Y}$ the map $|\mathcal{X}| \to |\mathcal{Y}|$ is continuous, and*
2. *for every morphism $U \to \mathcal{X}$ which is flat and locally of finite presentation with U an algebraic space the map of topological spaces $|U| \to |\mathcal{X}|$ is continuous and open.*

Proof. Choose a morphism $p : U \to \mathcal{X}$ which is surjective, flat, and locally of finite presentation with U an algebraic space. Such exist by the definition of an algebraic stack, as a smooth morphism is flat and locally of finite presentation (see Morphisms of Spaces, Lemmas 04AJ and 04TA). We define a topology on $|\mathcal{X}|$ by the rule: $W \subset |\mathcal{X}|$ is open if and only if $|p|^{-1}(W)$ is open in $|U|$. To show that this is independent of the choice of p, let $p' : U' \to \mathcal{X}$ be another morphism which is surjective, flat, locally of finite presentation from an algebraic space to $\mathcal{X}$. Set $U'' = U \times_{\mathcal{X}} U'$ so that we have a 2-commutative diagram

$$\begin{array}{ccc} U'' & \longrightarrow & U' \\ \downarrow & & \downarrow \\ U & \longrightarrow & \mathcal{X} \end{array}$$

As $U \to \mathcal{X}$ and $U' \to \mathcal{X}$ are surjective, flat, locally of finite presentation we see that $U'' \to U'$ and $U'' \to U$ are surjective, flat and locally of finite presentation, see Lemma lemma-property-spaces-too. Hence the maps $|U''| \to |U'|$ and $|U''| \to |U|$ are continuous, open and surjective, see Morphisms of Spaces, Definition 03ME and Lemma 042S. This clearly implies that our definition is independent of the choice of $p : U \to \mathcal{X}$.

Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. By Algebraic Stacks, Lemma 04T1 we can find a 2-commutative diagram

$$\begin{array}{ccc} U & \xrightarrow[a]{} & V \\ {\scriptstyle x}\downarrow & & \downarrow{\scriptstyle y} \\ \mathcal{X} & \xrightarrow{f} & \mathcal{Y} \end{array}$$

with surjective smooth vertical arrows. Consider the associated commutative diagram

$$\begin{array}{ccc} |U| & \xrightarrow[|a|]{} & |V| \\ {\scriptstyle |x|}\downarrow & & \downarrow{\scriptstyle |y|} \\ |\mathcal{X}| & \xrightarrow{|f|} & |\mathcal{Y}| \end{array}$$

of sets. If $W \subset |\mathcal{Y}|$ is open, then by the definition above this means exactly that $|y|^{-1}(W)$ is open in $|V|$. Since $|a|$ is continuous we conclude that $|a|^{-1}|y|^{-1}(W) = |x|^{-1}|f|^{-1}(W)$ is open in $|W|$ which means by definition that $|f|^{-1}(W)$ is open in $|\mathcal{X}|$. Thus $|f|$ is continuous.

Finally, we have to show that if U is an algebraic space, and $U \to \mathcal{X}$ is flat and locally of finite presentation, then $|U| \to |\mathcal{X}|$ is open. Let $V \to \mathcal{X}$ be surjective, flat, and locally of finite presentation with V an algebraic space. Consider the commutative diagram

$$\begin{array}{ccccc} |U \times_{\mathcal{X}} V| & \xrightarrow[e]{} & |U| \times_{|\mathcal{X}|} |V| & \xrightarrow[d]{} & |V| \\ & {\scriptstyle f}\searrow & {\scriptstyle c}\downarrow & & \downarrow{\scriptstyle b} \\ & & |U| & \xrightarrow{a} & |\mathcal{X}| \end{array}$$

Now the morphism $U \times_{\mathcal{X}} V \to U$ is surjective, i.e, $f : |U \times_{\mathcal{X}} V| \to |U|$ is surjective. The left top horizontal arrow is surjective, see Lemma lemma-points-cartesian. The morphism $U \times_{\mathcal{X}} V \to V$ is flat and locally of finite presentation, hence $d \circ e : |U \times_{\mathcal{X}} V| \to |V|$ is open, see Morphisms of Spaces, Lemma 042S. Pick $W \subset |U|$ open. The properties above imply that $b^{-1}(a(W)) = (d \circ e)(f^{-1}(W))$ is open, which by construction means that $a(W)$ is open as desired. $\square$